

# ASHRAE Standard 55-2023: Thermal Environmental Conditions for Human Occupancy

## Section 5.2 - Operative Temperature Bounds

The acceptable summer operative temperature range for building occupants wearing standard summer clothing of 0.5 clo is specified between 73.0 deg F and 79.0 deg F (22.8 deg C to 26.1 deg C).

Thermal comfort criteria require that operative temperatures remain strictly within the 80% acceptability limits under sedentary and light office activity.

## Section 5.3 - High Ambient Thermal Stress & Pre-cooling

When ambient urban micro-climate temperatures exceed 79.0 deg F (26.1 deg C), mechanical cooling systems must be modulated.

During extreme heat wave events exceeding 98.0 deg F to 105.0 deg F, active stage-2 or stage-3 HVAC pre-cooling must be initiated to maintain internal setpoints and reduce peak grid strain.

## Section 5.4 - Air Movement & Humidity Interaction

Elevated air velocity can provide occupant comfort at slightly higher operative temperatures provided relative humidity is controlled between 30% and 60%.